

ANALYSIS_SOURCES

Analysis Sources - Files Examined

This document lists all the source files examined during the creation of the comprehensive example projects analysis.

Qwen Code Sources

Documentation

- `cli_agents/qwen-code/README.md` - Main project overview
- `cli_agents/qwen-code/CHANGELOG.md` - Version history
- `cli_agents/qwen-code/docs/cli/authentication.md` - Auth details
- `cli_agents/qwen-code/docs/cli/commands.md` - Command reference
- `cli_agents/qwen-code/docs/cli/index.md` - CLI guide
- `cli_agents/qwen-code/docs/tools/` - Tools documentation

Source Code

- `cli_agents/qwen-code/package.json` - Dependencies and scripts
- `cli_agents/qwen-code/packages/cli/src/ui/models/availableModels.ts` - Model definitions
- `cli_agents/qwen-code/packages/core/src/core/openaiContentGenerator/provider/dashscope.ts` - DashScope provider implementation
- `cli_agents/qwen-code/packages/core/src/core/openaiContentGenerator/provider/` - Other provider implementations

Key Findings

- Monorepo architecture with core and CLI packages
 - Vision model auto-switching feature
 - DashScope cache control system
 - Session token management
 - Multi-authentication support (OAuth + API Key)
-

Gemini CLI Sources

Documentation

- `cli_agents/gemini-cli/README.md` - Main project overview
- `cli_agents/gemini-cli/GEMINI.md` - Development guidelines
- `cli_agents/gemini-cli/docs/get-started/authentication.md` - Auth setup
- `cli_agents/gemini-cli/docs/cli/checkpointing.md` - Session saving
- `cli_agents/gemini-cli/docs/cli/token-caching.md` - Token optimization
- `cli_agents/gemini-cli/docs/core/policy-engine.md` - Policy management
- `cli_agents/gemini-cli/docs/tools/mcp-server.md` - MCP integration

Source Code

- `cli_agents/gemini-cli/package.json` - Project structure and dependencies
- `cli_agents/gemini-cli/packages/core/src/config/models.ts` - Model definitions and fallback
- `cli_agents/gemini-cli/packages/core/src/core/client.ts` - GeminiClient implementation
- `cli_agents/gemini-cli/packages/core/src/core/geminiRequest.ts` - Request building
- `cli_agents/gemini-cli/packages/core/src/core/baseLlmClient.ts` - Base LLM client
- `cli_agents/gemini-cli/packages/core/src/code_assist/` - OAuth and credential handling
- `cli_agents/gemini-cli/packages/core/src/mcp/` - MCP provider implementations

Key Findings

- Monorepo with core, CLI, and a2a-server packages
 - Three authentication methods (OAuth, API Key, Vertex AI)
 - Thinking mode with token limits
 - Model fallback logic for service degradation
 - Loop detection and chat compression services
 - Comprehensive IDE context tracking
 - Three-tier release management (nightly, preview, stable)
-

DeepSeek CLI Sources

Documentation

- `cli_agents/deepseek-cli/README.md` - Main project overview
- `cli_agents/deepseek-cli/LOCAL-SETUP.md` - Local setup guide

Source Code

- `cli_agents/deepseek-cli/package.json` - Minimal dependencies
- `cli_agents/deepseek-cli/src/api.ts` - Dual-mode API client
- `cli_agents/deepseek-cli/src/cli.ts` - CLI interface
- `cli_agents/deepseek-cli/src/config.ts` - Configuration management
- `cli_agents/deepseek-cli/src/commands/interactive.ts` - Interactive REPL
- `cli_agents/deepseek-cli/src/commands/chat.ts` - Chat command
- `cli_agents/deepseek-cli/src/commands/setup.ts` - Setup automation
- `cli_agents/deepseek-cli/src/utils/exec.ts` - Execution utilities

Key Findings

- Single-package simple architecture
 - Dual local/cloud execution modes
 - Ollama integration
 - Minimal dependency footprint (5 core)
 - Setup automation and health checking
 - Response formatting with markdown support
 - Model availability detection
-

Analysis Methodology

Approach

1. Examined README files for high-level overview
2. Reviewed package.json for dependencies and structure
3. Analyzed key source files for implementation patterns
4. Reviewed documentation for features and capabilities
5. Identified unique features and best practices
6. Compared approaches across three projects

Artifacts Created

1. `EXAMPLE_PROJECTS_ANALYSIS.md` - Comprehensive 680-line analysis

- 2. EXAMPLE_PROJECTS_QUICK_REFERENCE.md - Quick lookup guide
 - 3. ANALYSIS_SOURCES.md - This file
-

Feature Extraction Summary

Vision & Multimodal (Qwen Code)

- Image detection in prompts
- Vision model switching modes (once, session, persist)
- Vision-specific API parameters
- High-resolution image support

Search & Real-Time Info (Gemini CLI)

- Google Search grounding
- Search-aware response generation
- Integration with Gemini models

Checkpointing & Session Management

- Qwen: Session token limits and compression
- Gemini: Full checkpoint save/restore
- DeepSeek: Basic session history

Authentication Patterns

- Qwen: OAuth + OpenAI-compatible fallback
- Gemini: OAuth → API Key → Vertex AI
- DeepSeek: Local Ollama → Cloud API

Performance Optimization

- Token caching (ephemeral)
- Chat compression
- Context window management
- Model fallback strategies

Tool Integration

- MCP Protocol (Qwen, Gemini)
 - File system operations
 - Shell execution
 - Web fetching and search
-

Key Code Locations for Reference

Provider Implementations

- Qwen DashScope: `Qwen_Code/packages/core/src/core/openaiContentGenerator/provider/dashscope.ts`
- Gemini Client: `Gemini_CLI/packages/core/src/core/client.ts`
- DeepSeek API: `DeepSeek_CLI/src/api.ts`

Authentication

- Qwen Auth: `Qwen_Code/packages/core/src/core/` (OAuth & credential storage)
- Gemini Auth: `Gemini_CLI/packages/core/src/code_assist/` and `packages/core/src/core/`
- DeepSeek Config: `DeepSeek_CLI/src/config.ts`

Models & Configuration

- Qwen Models: `Qwen_Code/packages/cli/src/ui/models/availableModels.ts`
- Gemini Models: `Gemini_CLI/packages/core/src/config/models.ts`
- DeepSeek Models: `DeepSeek_CLI/src/config.ts`

Advanced Features

- Vision Switching: `Qwen_Code/packages/cli/src/ui/commands/modelCommand.ts`
 - Checkpointing: `Gemini_CLI/packages/core/src/` (ChatCompressionService)
 - Setup Automation: `DeepSeek_CLI/src/commands/setup.ts`
-

Recommendations for HelixCode Integration

Phase 1: Foundation

1. Import DeepSeek provider pattern (simplest)
2. Add Qwen OAuth authentication
3. Implement Gemini API key support

Phase 2: Advanced Features

1. Implement vision model auto-switching

2. Add token caching strategies
3. Implement model fallback logic

Phase 3: Integration

1. Add MCP protocol support
2. Integrate Google Search grounding
3. Implement checkpointing system

Phase 4: Enterprise

1. Add loop detection
2. Implement chat compression
3. Add IDE context tracking

Document Status

- **Analysis Date:** November 6, 2025
- **Projects Analyzed:** 3
- **Total Files Examined:** 50+
- **Documentation Generated:** 3 markdown files
- **Total Lines of Analysis:** 1000+

All analysis documents are stored in `/Users/milosvasic/Projects/helix_code/`